

A Technical Introduction to Time series forecasting

Section 1

History Jevons was the first to study times series. In his book he examined a number of weekly financial time series between the years 1825 to 1860 inclusive. These included times series of bankruptcies, currency circulation and discount rates.

Section 2

In general, a function approximation problem asks us to select a function among a well-defined class that closely matches ("approximates") a target function in a task-specific way. One can distinguish two major classes of function approximation problems: First, for known target functions, approximation theory is the branch of numerical analysis that investigates how certain known functions (for example, special functions) can be approximated by a specific class of functions (for example, polynomials or rational functions) that often have desirable properties (inexpensive computation, continuity, integral and limit values, etc.). Second, the target function, call it g , may be unknown; instead of an explicit formula, only a set of points (a time series) of the form $(x, g(x))$ is provided. Depending on the structure of the domain and codomain of g , several techniques for approximating g may be applicable. For example, if g is an operation on the real numbers, techniques of interpolation, extrapolation, regression analysis, and curve fitting can be used. If the codomain (range or target set) of g is a finite set, one is dealing with a classification problem instead. A related problem of online time series approximation is to summarize the data in one-pass and construct an approximate representation that can support a variety of time series queries with bounds on worst-case error. To some extent, the different problems (regression, classification, fitness approximation) have received a unified treatment in statistical learning theory, where they are viewed as supervised learning problems.

Section 3

Additionally, time series analysis techniques may be divided into parametric and non-parametric methods. The parametric approaches assume that the underlying stationary stochastic process has a certain structure which can be described using a small number of parameters (for example, using an autoregressive or moving-average model). In these approaches, the task is to estimate the parameters of the model that describes the stochastic process. By contrast, non-parametric approaches explicitly estimate the covariance or the spectrum of the process without assuming that the process has any particular structure. Methods of time series analysis may also be divided into linear and non-linear, and univariate and

multivariate.

Section 4

Subsequence time series clustering Subsequence time series clustering resulted in unstable (random) clusters induced by the feature extraction using chunking with sliding windows. It was found that the cluster centers (the average of the time series in a cluster - also a time series) follow an arbitrarily shifted sine pattern (regardless of the dataset, even on realizations of a random walk). This means that the found cluster centers are non-descriptive for the dataset because the cluster centers are always nonrepresentative sine waves.

Section 5

Curve fitting is the process of constructing a curve, or mathematical function, that has the best fit to a series of data points, possibly subject to constraints. Curve fitting can involve either interpolation, where an exact fit to the data is required, or smoothing, in which a "smooth" function is constructed that approximately fits the data. A related topic is regression analysis, which focuses more on questions of statistical inference such as how much uncertainty is present in a curve that is fit to data observed with random errors. Fitted curves can be used as an aid for data visualization, to infer values of a function where no data are available, and to summarize the relationships among two or more variables. Extrapolation refers to the use of a fitted curve beyond the range of the observed data, and is subject to a degree of uncertainty since it may reflect the method used to construct the curve as much as it reflects the observed data.

Section 6

Univariate linear measures Moment (mathematics) Spectral band power Spectral edge frequency Accumulated energy (signal processing) Characteristics of the autocorrelation function Hjorth parameters FFT parameters Autoregressive model parameters Mann–Kendall test Univariate non-linear measures Measures based on the correlation sum Correlation dimension Correlation integral Correlation density Correlation entropy Approximate entropy Sample entropy Fourier entropy Wavelet entropy Dispersion entropy Fluctuation dispersion entropy Rényi entropy Higher-order methods Marginal predictability Dynamical similarity index State space dissimilarity measures Lyapunov exponent Permutation methods Local flow Other univariate measures Algorithmic complexity Kolmogorov complexity estimates Hidden Markov model states Rough path signature Surrogate time series and surrogate correction Loss of recurrence (degree of non-stationarity) Bivariate linear measures Maximum linear cross-correlation Linear Coherence (signal processing) Bivariate non-linear measures Non-linear interdependence Dynamical

Entrainment (physics) Measures for phase synchronization
Measures for phase locking Similarity measures:
Cross-correlation Dynamic time warping Hidden Markov
model Edit distance Total correlation Newey–West
estimator Prais–Winsten transformation Data as vectors in
a metrizable space Minkowski distance Mahalanobis
distance Data as time series with envelopes Global
standard deviation Local standard deviation Windowed
standard deviation Data interpreted as stochastic series
Pearson product-moment correlation coefficient
Spearman's rank correlation coefficient Data interpreted as
a probability distribution function Kolmogorov–Smirnov test
Cramér–von Mises criterion

models, the acronyms are extended with a final "X" for
"exogenous".

Section 7

Splitting a time-series into a sequence of segments. It is
often the case that a time-series can be represented as a
sequence of individual segments, each with its own
characteristic properties. For example, the audio signal
from a conference call can be partitioned into pieces
corresponding to the times during which each person was
speaking. In time-series segmentation, the goal is to
identify the segment boundary points in the time-series,
and to characterize the dynamical properties associated
with each segment. One can approach this problem using
change-point detection, or by modeling the time-series as a
more sophisticated system, such as a Markov jump linear
system.

Section 8

Classical models (AR, ARMA, ARIMA, and well-known
variations) Models for time series data can have many
forms and represent different stochastic processes. When
modeling variations in the level of a process, three broad
classes of practical importance are the autoregressive (AR)
models, the integrated (I) models, and the moving-average
(MA) models. These three classes depend linearly on
previous data points. Combinations of these ideas produce
autoregressive moving-average (ARMA) and
autoregressive integrated moving-average (ARIMA)
models. The autoregressive fractionally integrated
moving-average (ARFIMA) model generalizes the former
three. Another important generalization is the time-varying
autoregressive (TVAR) model, in which the AR coefficients
are allowed to change over time, enabling the model to
capture evolving or non-stationary dynamics. Extensions of
these classes to deal with vector-valued data are available
under the heading of multivariate time-series models and
sometimes the preceding acronyms are extended by
including an initial "V" for "vector", as in VAR for vector
autoregression. An additional set of extensions of these
models is available for use where the observed time-series
is driven by some "forcing" time-series (which may not
have a causal effect on the observed series): the distinction
from the multivariate case is that the forcing series may be
deterministic or under the experimenter's control. For these